

Drug News and FDA Alerts

Drug News and FDA Alerts by Year

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2021 FDA Alerts ▼

- **October 29, 2021: Zimmer Biomet ROSA One 3.1 Brain Application–Class I Recall due to Software Error** - Zimmer Biomet is recalling the ROSA One 3.1 Brain Application due to a software error that could lead to incorrect placement of instruments during stereotactic neurosurgical procedures (for example, techniques to direct the tip of a tool using coordinates provided by medical imaging to reach a specific part of the brain). If such an error occurs, it could cause adverse events such as stroke, serious injury, severe disability, and death. There have been three complaints and no deaths or injuries reported about this device issue. The FDA has identified this as a Class I recall, the most serious type of recall. Use of these devices may cause serious injuries or death. For further details, see the [FDA recall notice](#). An [FDA recall database entry](#) provides guidance on avoiding steps that may trigger the software error and actions to be taken without delay by all users of the ROSA One 3.1 Brain application device. On September 22, 2021, Zimmer Biomet alerted affected consignees of the software via “Urgent Medical Device Correction” letters. The company expects to visit each customer site, estimated between February and May 2022, to implement software updates. Both the recall database entry and the recall notice provide contact information.

- **October 28, 2021: Ethosuximide (Zarontin®)** – FDA approved an update to the Warnings and Adverse Reactions sections of the [prescribing information](#) to add thrombocytopenia, as described in the [approval letter](#).
- **October 14, 2021: Lacosamide (Vimpat®)** - FDA expanded age indications for oral and intravenous monotherapy and adjunctive therapy in the treatment of partial onset seizure patients, to include ages ≥ 1 month to < 4 years (previously ≥ 4 years). Age indications remain at ≥ 4 years for use as adjunctive therapy in the treatment of primary generalized tonic-clonic seizures. Details and updated dosage and oral solution shelf-life information are found in the [prescribing information](#) and associated [FDA approval letter](#).
- **September 30, 2021: Cannabidiol (Epidiolex®)** - As summarized in this [FDA Drug Safety-related labeling alert](#), with full details in the [prescribing information](#), FDA announced label changes related to:
 - *Drug Interactions (Section 7.2, Effect of Epidiolex on Other Drugs)* – New information about coadministration of Epidiolex with sensitive P-gp substrates given orally:
 - **Everolimus:** When initiating Epidiolex in patients taking everolimus, monitor therapeutic drug levels of everolimus and adjust the dosage accordingly. When initiating everolimus in patients taking a stable dosage of Epidiolex, a lower starting dose of everolimus is recommended, with therapeutic drug monitoring.
 - **Other orally administered sensitive P-gp substrates** (e.g., sirolimus, tacrolimus, digoxin): Increases in exposure may be observed on coadministration with Epidiolex, so therapeutic drug monitoring and dose reduction of other P-gp substrates should be considered when given orally and concurrently with Epidiolex.
 - *Adverse Reactions (Section 6.1, Clinical Trials Experience)* – *Increases in pneumonia with concomitant clobazam:* This added subsection reports clinical trial observations of increases in pneumonia in patients with Lennox-Gastaut syndrome (LGS), Dravet syndrome (DS), and tuberous sclerosis complex (TSC) who were receiving Epidiolex with concomitant clobazam.
- **September 15, 2021: Lacosamide (Vimpat®)** - Dyskinesia is added to neurologic disorders in Subsection 6.2 (Adverse Reactions;

Postmarketing Experience) of the [prescribing information](#), to read "Neurologic disorders: dyskinesia, new or worsening seizures."

- **August 27, 2021: Brivaracetam (Briviact®)** – FDA approved expanded age indications for Briviact for the treatment of partial-onset seizures to include patients one month of age and older, applicable to brivaracetam tablets for oral use, oral solution, and injection for intravenous use. The [PI](#) and [approval letter](#) provide details.
- **August 18, 2021: Methsuximide (Celontin®)** - An anticipated temporary shortage in August-September 2021 is reported by Parke-Davis, division of Pfizer Inc. As described in the [2010 Celontin Prescribing Information](#) (PI), this drug was approved for the treatment of absence seizures refractory to other drugs. Currently, there is no generic form of methsuximide available. Methsuximide is in the succinimide class of antiseizure medications (ASMs) along with ethosuximide ([2016 PI](#)) and phensuximide. ([Chen et al. 1963. PMID 14020499](#)) Adverse effects of the succinimide ASMs are broadly similar. In some circumstances, if a patient's methsuximide dosing could be interrupted during a shortage, providers may wish to consider switching to ethosuximide (Zarontin®). For more information about ethosuximide, see the [AES 2020 summary of ASMs available in the United States](#). As of August 18, 2021, a methsuximide shortage is not reported in the [FDA Drug Shortages database](#). AES will continue to monitor and alert members to status updates.
- **June 30, 2021: Topiramate (Topamax®)** label change. The FDA approved a label change to add the following ophthalmologic findings to subsection 5.1 (Warnings and Precautions; Acute Myopia and Secondary Angle Closure Glaucoma Syndrome): choroidal detachments, retinal pigment and epithelial detachments, and macular striae. Details are available in the [prescribing information](#) and [approval letter](#).
- **April 16, 2021: Everolimus (Afinitor® and Afinitor/Disperz®)** tablets for oral suspension – The FDA updated [Prescribing Information](#) to include a new Warning and Precaution: Radiation sensitization and recall, in some cases severe, involving cutaneous and visceral organs (including radiation esophagitis and pneumonitis) have

been reported in patients treated with radiation prior to, during, or subsequent to Afinitor/Afinitor Disperz treatment. Updates include advice to monitor patients closely when the drug is administered during or sequentially with radiation treatment. Corresponding updates were made to PI sections on Adverse Reactions, Postmarketing Experience, Patient Counseling information, and Patient Information/Patient Package Insert.

- **March 31, 2021: Lamotrigine (Lamictal®)** – FDA updated [Prescribing Information](#) Warnings and Precautions summary related to cardiac rhythm and conduction abnormalities: “Based on in vitro findings, Lamictal could cause serious arrhythmias and/or death in patients with certain underlying cardiac disorders or arrhythmias. Any expected or observed benefit of Lamictal in an individual patient with clinically important structural or functional heart disease must be carefully weighed against the risk for serious arrhythmias and/or death for that patient.” This replaces the summary from an [October 2020 PI update](#) that read, “Avoid Lamictal in patients with underlying cardiac disorders or arrhythmias.”

Along with the March 31, 2021 PI update, FDA issued and broadly distributed a [Drug Safety Communication](#) about lamotrigine. The communication summarized related concerns and announced that FDA will require postmarket safety studies to evaluate whether other sodium channel blockers in the same drug class, as listed in the communication, have similar effects on the heart. The communication advises, “...sodium channel blockers approved for epilepsy, bipolar disorder, and other indications should not be considered safer alternatives to lamotrigine in the absence of additional information.” Specific guidance for health care professionals is provided as well patient messaging that states in part, “Patients should not stop taking your medicine without first talking to your prescriber, because stopping lamotrigine can lead to uncontrolled seizures, or new or worsening mental health problems...”

AES provided practical guidance for clinicians through a recorded March 11 webinar, [Lamotrigine and the Heart, Cause for Concern?](#)

and a joint [ILAE/AES advisory statement on cardiac effects of lamotrigine in Epilepsy Currents](#). AES will continue to keep members apprised of related news.

- **February 16, 2021: Phenytoin (Dilantin-125®)** – FDA updated the [Prescribing Information](#) related to serious dermatologic reactions (severe cutaneous adverse reactions, or SCARs), drug interactions, and use in patients with decreased CYP2C9 function. Updates include the following additions to specific PI sections. Please refer to the [full PI](#) for context.

Section 5.3, Serious Dermatologic Reactions *(additions bolded below)*

...In addition, retrospective, case-control, genome-wide association studies in patients of southeast Asian ancestry have also identified an increased risk of SCARs in carriers of the decreased function CYP2C9*3 variant, which has also been associated with decreased clearance of phenytoin. **Consider avoiding DILANTIN as an alternative to carbamazepine in patients who are positive for HLA-B*1502 or in CYP2C9*3 carriers [see Use in Specific Populations (8.7) and Clinical Pharmacology (12.5)].**

The use of HLA-B*1502 or CYP2C9 genotyping has important limitations and must never substitute for appropriate clinical vigilance and patient management. The role of other possible factors in the development of, and morbidity from, SJS/TEN, such as antiepileptic drug (AED) dose, compliance, concomitant medications, comorbidities, and the level of dermatologic monitoring have not been studied.

Section 7, Drug Interactions *(additions underlined)*

Phenytoin is extensively bound to plasma proteins and is prone to competitive displacement. Phenytoin is **primarily** metabolized by the hepatic cytochrome P450 enzyme CYP2C9 and **to a lesser extent by** CYP2C19 and is particularly susceptible to inhibitory drug interactions because it is subject to saturable metabolism. ...

Section 8, Use in Specific Populations, Subsection 8.7 Use in Patients with Decreased CYP2C9 Function (*new subsection added*)

Patients who are intermediate or poor metabolizers of CYP2C9 substrates (e.g., *1/*3, *2/*2, *3/*3) may exhibit increased phenytoin serum concentrations compared to patients who are normal metabolizers (e.g., *1/*1). Thus, patients who are known to be intermediate or poor metabolizers may ultimately require lower doses of phenytoin to maintain similar steady-state concentrations compared to normal metabolizers. If early signs of dose-related central nervous system (CNS) toxicity develop, serum concentrations should be checked immediately [*see Clinical Pharmacology (12.5)*].

- **February 5, 2021: Benzodiazepines** – clonazepam, diazepam, midazolam, lorazepam, and clobazam (various manufacturers and formulations); FDA announced updates to PI for benzodiazepines, including these used as antiseizure medications. Among the black-box safety warnings now included are:
 - Concomitant use with opioids may result in profound sedation, respiratory depression, coma, and death
 - Risk of abuse, misuse, and addiction, which can lead to overdose and death
 - Abrupt discontinuation or rapid dosage reduction after continued use may precipitate acute withdrawal reactions,

which can be life-threatening

As just one example, clobazam changes are detailed, with prominent black box warnings, in [its PI](#) and summarized in the corresponding [label revision approval letter](#). Each drug can be searched in the [Drugs@FDA database](#), with links to the PI and letter found for each specific manufacturer and formulation under the header, Approval Date(s) and History, Letters, Labels, Reviews.

- **January 29, 2021:** FDA Safety Warning on the Cardiac Effects of **Lamotrigine**: [An Advisory from the Ad Hoc ILAE/AES Task Force](#) – A preprint version of this joint ILAE/AES statement is provided here in advance of copublication in Epilepsia Open and Epilepsy Currents to assist AES members in managing clinical care and communications with patients about the October 9, 2020, new safety warning and [prescribing information](#) updates related to lamotrigine.

2020 FDA Alerts



2019 FDA Alerts



- 12/20/19 - The U.S. Food and Drug Administration (FDA) announced a recall of the LivaNova VNS Therapy SenTiva Generator due to an unintended reset error that causes the system to stop delivering VNS therapy. If device replacement is needed, there is a risk associated with additional surgery to replace the generator. LivaNova has received 14 reports of unexpected reset errors. 4 patients have required early revision surgery for failed devices. No deaths related to this issue have been reported. On July 31, 2019, LivaNova implemented additional mitigations and at this time, no reset errors have been observed since the implementation of these mitigations. These additional mitigations are currently under review by the FDA. The [full FDA announcement](#) provides details and recommended actions for providers and patients.
- 12/19/19 - The U.S. Food & Drug Administration (FDA) issued a [Drug Safety Communication](#) warning that serious breathing difficulties may occur in patients using gabapentin (Neurontin, Gralise, Horizant) or pregabalin (Lyrica, Lyrica CR) who have respiratory risk factors. These include use of opioid pain medicines and other drugs

that depress the central nervous system and conditions such as chronic obstructive pulmonary disease (COPD) that reduce lung function. The elderly are also at higher risk. An [FDA portal on the safety communication](#) links to additional details, including data on which the safety concern is based, advice for health care professionals and patients and caregivers, and where to report side effects. In addition to the safety communication, FDA is requiring that 1) new warnings about the risk of respiratory depression be added to the prescribing information of the gabapentinoids, and 2) drug manufacturers conduct clinical trials to further evaluate abuse potential of the gabapentinoids, particularly in combination with opioids, because misuse and abuse of these products together is increasing, and co-use may increase the risk of respiratory depression. Gabapentin and pregabalin are FDA-approved for a variety of conditions, including seizures, nerve pain, and restless legs syndrome. [FDA announced in July 2019](#) the approval of the first pregabalin generics of Lyrica.

- 11/21/19 – The U.S. Food and Drug Administration (FDA) approved **cenobamate** tablets (Xcopri, SK Life Science Inc.) as a new treatment option for partial onset seizures in adults, based on two randomized controlled trials that established safety and efficacy. The [FDA announcement](#) was released November 21, and [highlights of prescribing Information \(PI\)](#) are dated 11/2019, pending DEA controlled substance scheduling. It is hoped that **cenobamate** (Xcopri) will be available in pharmacies shortly after scheduling has been completed.
- 10/23/19 – The FDA approved a prescribing information (PI) change for Keppra® (levetiracetam). Previously approved only for adjunctive therapy, Keppra indications now include monotherapy for partial-onset seizures in patients 1 month of age and older. The updated PI includes related updates to information on dosing and discontinuation. Recommended dosing is the same for monotherapy and adjunctive therapy. The updated PI includes no change to Keppra indications for use only as adjunctive therapy in the treatment of myoclonic seizures in patients 12 years of age and older with juvenile myoclonic epilepsy (JME) and in the treatment of primary generalized tonic-clonic seizures (GTCS) in patients 6 years

of age and older with idiopathic generalized epilepsy. See highlights of the [updated PI](#).

- 7/31/19 – The FDA approved addition of the following statement to section 5.4 (*Warnings and Precautions; Neurotoxicity*) of the Prescribing Information (PI) for Sabril (vigabatrin): "Intramyelinic edema (IME) has been reported in postmortem examination of infants being treated for IS with vigabatrin." See details in the [FDA letter of approval](#) and the [updated PI](#).
- 7/22/19 - A July 22, 2019 [announcement from the U.S. Food and Drug Administration, Center for Drug Evaluation and Research \(FDA CDER\)](#) highlighted the July 19, 2019 approval of multiple applications of first generics of Lyrica (pregabalin) for indications that include "adjunctive therapy for the treatment of partial onset seizures in patients 17 years of age and older." CDER Director, Janet Woodcock, MD, cited "FDA's longstanding commitment to advance patient access to lower cost, high-quality generic medicines." The announcement also summarizes patient Medication Guide warnings required to be dispensed with pregabalin. Details (dosage levels, manufacturers, and indications) about first-time generic drugs approved in 2019, including pregabalin, can be found at this [FDA First Generic Drug Approvals 2019 page](#).
- 5/17/19 - On May 17, 2019, the Food & Drug Administration approved Nayzilam (midazolam) intranasal spray for the treatment of seizure clusters and acute repetitive seizures that are distinct from the patient's usual seizure pattern in patients with epilepsy 12 years of age and older. The prescribing information may be found [here](#).
- 2/26/19 Intravenous Levetiracetam distributed by Dr. Reddy's Laboratories Inc. A labeling error incorrectly stating the product strength resulted in a Class I recall of 2770 i.v. bags of levetiracetam in 0.54% sodium chloride injection, 1500 mg/100 mL (15 mg/mL), distributed in the US by Dr. Reddy's Laboratories Inc. The February 20, 2019, FDA Enforcement Report states, "the preprinted text on the primary infusion bag and the NDC incorrectly identifies the product as levetiracetam in 0.75% sodium chloride (1000 mg/100 mL). However, the external foil pouch correctly identifies the product as levetiracetam in 0.54% sodium chloride injection (1500

mg/100 mL)." The recall affects bags manufactured by Gland Pharma Limited from lot ABD807 (Exp. 5/20). More [here](#).

- 2/26/19 - Lamictal (lamotrigine): Health Canada and the European Medicine Agency have recently issued 2 warnings regarding lamotrigine. First, Brugada-type ECG arrhythmogenic ST-T abnormalities and typical Brugada ECG pattern have been reported in patients treated with lamotrigine. The use of lamotrigine should be carefully considered in patients with Brugada syndrome. Second, lamotrigine also has been reported to pass into breast milk, resulting in lamotrigine plasma total levels in neonates and older infants of up to approximately 50% of the mother's level. Therefore, in some breastfed infants, plasma concentrations of lamotrigine may reach levels at which pharmacological effects occur. The potential benefits of breastfeeding should be weighed against the potential risk of adverse effects occurring in the infant. Should a woman decide to breastfeed while on therapy with lamotrigine, the infant should be monitored for adverse effects, such as sedation, rash, and poor weight gain. More [here](#).
- 1/17/19 - The FDA has cleared the Embrace watch for use in epilepsy patients ages 6 years and older. It was first approved in January 2018 for adults aged ≥ 21 years. The watch detects convulsive seizures by combining information from both electrodermal and accelerometry biosensors. When a seizure is detected, an automated alert can be sent to a caregiver, when the watch detects unusual patterns that may be associated with a convulsive seizure. Clinical trials have demonstrated accuracy rates greater than 95% for detecting generalized convulsive seizures. More [here](#).

2018 FDA Alerts

- 11/11/18 - The FDA approved new prescribing information for Vimpat (lacosamide). In addition to the previously known 0.4% risk of first-degree AV block (PR interval >200 msec) observed in preclinical trials, additional cardiac risks have been identified. "In the post-marketing setting, there have been reports of cardiac arrhythmias in patients treated with VIMPAT, including bradycardia,

AV block, and ventricular tachyarrhythmia, which have rarely resulted in asystole, cardiac arrest, and death. Most, although not all, cases have occurred in patients with underlying proarrhythmic conditions, or in those taking concomitant medications that affect cardiac conduction or prolong the PR interval. These events have occurred with both oral and intravenous routes of administration and at prescribed doses as well as in the setting of overdose. Vimpat should be used with caution in patients with underlying proarrhythmic conditions such as known cardiac conduction problems (e.g., marked first-degree AV block, second-degree or higher AV block, and sick sinus syndrome without pacemaker), severe cardiac disease (such as myocardial ischemia or heart failure, or structural heart disease), and cardiac sodium channelopathies (e.g., Brugada Syndrome). VIMPAT should also be used with caution in patients on concomitant medications that affect cardiac conduction, including sodium channel blockers, beta-blockers, calcium channel blockers, potassium channel blockers, and medications that prolong the PR interval. The PI PDF is attached [here](#).

- 11/12/18 - The FDA approved new prescribing information (PI) for VIMPAT[BOLD], which now contains additional serious warnings about cardiac arrhythmias including ventricular arrhythmias. VIMPAT is indicated for the treatment of partial-onset seizures in patients 4 years of age and older. Additional information regarding the updated PI is located [here](#).
- 10/18/18 - The FDA approved new labeling for oral Dilantin [BOLD] (phenytoin) suspension. Section 5.6 of the Prescribing information indicates, "Cases of bradycardia and cardiac arrest have been reported in Dilantin-treated patients, both at recommended phenytoin doses and levels and in association with phenytoin toxicity. Most of the reports of cardiac arrest occurred in patients with underlying cardiac disease." It is important not to confuse this warning with the existing concerns for intravenous phenytoin or fosphenytoin. The updated labeling information is located [here](#).
- 9/1/2018 - The FDA has extended the labeling for Fycompa (perampanel). It is now indicated for the treatment of focal onset seizures with or without progression to bilateral tonic-clonic

seizures in children as young as age 4 years. This indication includes both monotherapy and adjunctive therapy. The revised PI can be located [here](#).

- 8/20/18 - The FDA approved Diacomit (stiripentol) for the treatment of seizures associated with Dravet syndrome in persons 2 years of age and older. Diacomit is approved for use as an adjunct to clobazam, with which it has a drug-drug interaction. It is available as capsules or oral powder for suspension and is not scheduled. Prescribing information is located on the FDA CDER website [here](#).
- 8/8/18 - The FDA has launched a new medication guide database to replace the current medication guide webpage. Features of the new database include active ingredient and brand name search for medication guides, download capabilities to CSV and Excel spreadsheets, streamlined data entry for faster database updates, and improved mobile device usability. Click [here](#) to view the new medication guide database.
- 7/18/18 - The FDA updated the prescribing information (PI) for Lamictal (lamotrigine) to include the extremely rare occurrence of hemophagocytic lymphohistiocytosis in 8 cases worldwide since 1994. The revised PI states, "Hemophagocytic lymphohistiocytosis (HLH) has occurred in pediatric and adult patients taking Lamictal for various indications. HLH is a life-threatening syndrome of pathologic immune activation characterized by clinical signs and symptoms of extreme systemic inflammation. It is associated with high mortality rates if not recognized early and treated. Common findings include fever, hepatosplenomegaly, rash, lymphadenopathy, neurologic symptoms, cytopenias, high serum ferritin, and liver function and coagulation abnormalities. In cases of HLH reported with Lamictal, patients have presented with signs of systemic inflammation (fever, rash, hepatosplenomegaly, and organ system dysfunction) and blood dyscrasias. Symptoms have been reported to occur within 8 to 24 days following the initiation of treatment." Providers should review the new PI which provides further guidance on differential diagnosis and treatment. HLH has signs which may overlap with other diagnoses including Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS),

Stevens-Johnson syndrome, and toxic epidermal necrolysis. The July 2018 revised PI can be located [here](#).

- 6/25/2018 - The FDA approved **Epidiolex** (cannabidiol) for the treatment of seizures associated with Lennox-Gastaut syndrome or Dravet syndrome in patients two years of age and older. This new antiepileptic drug will be available shortly after the Drug Enforcement Agency issues a scheduling determination 90 days after the FDA-approval date. Additional information is located [here](#) and the [PI is here](#).
- 5/10/2018 - The FDA approved an extension to the indication for **Briviact** (brivaracetam). Briviact is now indicated for the treatment of partial onset seizures in children four years of age and older, and in adults. The i.v. formulation is not approved for children ages four to 15 years. Additional information for download is located [here](#).
- 4/27/2018 - The FDA approved the **Medtronic Deep Brain Stimulation (DBS) System for Epilepsy**. The device delivers electrical pulses to a location inside the brain which is involved in seizures. The system consists of a pulse generator (IPG) implanted under the skin of the upper chest, and two leads implanted in the brain. The Medtronic DBS System for Epilepsy helps reduce the frequency of seizures in epilepsy patients who have frequent, disabling, partial-onset seizures and have not responded well to antiepileptic medications. Additional information is located [here](#).
- 4/25/2018 The FDA is providing health care providers with preliminary information concerning **magnetic resonance (MR) thermometry** reliability with **magnetic resonance-guided laser interstitial thermal therapy (MRgLITT)** devices.

The FDA is currently evaluating data, which suggests that potentially inaccurate MR thermometry information can be displayed during treatment. For example, MR parameters such as voxel size (measurement of the image resolution or detail) and MR image acquisition time (e.g., up to 8 seconds) may contribute to inaccurate MR thermometry readings and potential errors in the ablation assessment. In addition, MRgLITT devices may not account for the continued thermal spread of energy to the surrounding

tissue (as the target ablation area returns to its baseline temperature), which may result in an underestimation of thermal damage.

The FDA reviewed Medical Device Reports (MDRs) and literature reports which describe adverse events, such as neurological deficits (e.g., focal motor deficits, aphasia, cognitive changes), increased intracerebral edema or pressure, intracranial bleeding, and/or visual changes (e.g., visual field deficits, blurry vision) when these devices were used to treat intra-cranial lesions. Several of these reports note which events required urgent medical and/or surgical intervention, and may have been associated with patient deaths. However, it is unclear at this time, whether an inaccuracy of MR thermometry directly caused or contributed to these events. Additional information is located [here](#).

- 4/25/2018 - The FDA has issued a new warning that the medication **Lamictal** (lamotrigine) for seizures and bipolar disorder can cause a rare, but very serious reaction that excessively activates the body's infection-fighting immune system. This can cause severe inflammation throughout the body and lead to hospitalization and death, especially if the reaction is not diagnosed and treated quickly. As a result, the FDA is requiring a new warning about this risk be added to the prescribing information in the lamotrigine drug labels. Additional information is located [here](#).
- 4/10/2018 - The FDA approved the use of **Afinitor** (everolimus) for the adjunctive treatment of adult and pediatric patients aged two years and older with tuberous sclerosis complex (TSC)-associated focal-onset seizures. Additional information is located [here](#).

2017 FDA Alerts

- 11/02/2017 - The FDA expanded the approval for **Vimpat** oral tablets and solution to include children. The approval is now for the treatment of partial-onset seizures in patients age 4 and older as monotherapy or adjunctive therapy. Vimpat injection remains approved for this indication only for patients 17 years and older. More information is available [here](#).

- 08/2017 - Health Canada updated the **Keppra** monograph to add warnings for pancytopenia and rhabdomyolysis and is available [here](#). The risk of pancytopenia has been included in the warnings and precautions sections of the Canadian product monograph for Keppra (levetiracetam). The risk of rhabdomyolysis/blood creatine phosphokinase increase has been included in the post-market adverse drug reactions section.
- 08/17/2017 - The FDA approved an extension of the indication for eslicarbazepine (**Aptiom**). Aptiom is now indicated down to age 4 years for both monotherapy and adjunctive therapy for the treatment of partial (focal) onset seizures. The updated prescribing information can be found [here](#).
- 08/15/2017 - The FDA approved an extension on the indication for **brivaracetam (Briviact)** Briviact is now indicated for both monotherapy and adjunctive therapy for persons age 16 years and above for the treatment of partial (focal) onset seizures. The updated prescribing information can be found [here](#).
- 07/26/2017 - The FDA approved an important change to the **Fycompa (perampanel)** prescribing information. Perampanel is now approved as monotherapy for patients 12 years of age and older who have partial (focal)-onset seizures both with and without progression to generalized tonic-clonic seizures. This new, expanded FDA indication does not include monotherapy for primary generalized tonic-clonic seizures, where perampanel is still approved as adjunctive treatment. This is the first antiepileptic drug (AED) to receive approval for monotherapy use under the FDA's new rule granting a monotherapy indication based solely on the results of randomized controlled trials of adjunctive therapy for partial (focal)-onset seizures. Additional information regarding the prescribing information is located [here](#).
- 03/28/2017 - The FDA announced an update to the **Vimpat** (lacosamide) prescribing information to include a warning about rare cases of Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS; formerly known as multiorgan hypersensitivity). Additionally, cardiogenic shock has been added as an effect of Vimpat overdose. More information is available [here](#).

- 03/07/2017 - The FDA sent a letter to Sunovion asking that Prescribing Information be updated to include the occurrence of toxic epidermal necrolysis (TEN) with the use of **Aptiom**. This is in addition to the warnings of Stevens-Johnson Syndrome (SJS) and Drug Reaction with Eosinophilia and Systemic Symptoms (DRWSS) listed in the current Prescribing Information. More [here](#).
- 02/20/2017 - VistaPharm, Inc is recalling more than 12,000 units of **Phenytoin Oral Suspension**, USP 100 mg/ 4 mL, 50 unit dose cups of 4 mL per case, Rx only, according to the US Food and Drug Administration's (FDA) latest [Drug Enforcement Report](#) (February 15, 2017). The recall was initiated on January 18, 2017, due to, 'CGMP Deviations: Purified water used to manufacture the drug products may have been contaminated with *Burkholderia cepacia*,' according to the FDA. The affected products were distributed throughout the United States. Known lot numbers include 428700 (Exp. 11/17) and 409500 (Exp. 06/17). On February 2, 2017, the FDA designated the recall a class II, meaning that exposure might cause temporary health problems, but is unlikely to cause injuries of a serious nature.
- 02/07/2017 - Health Canada has published a review that concluded that there may be a link between the use of **levetiracetam** (Keppra) and the risk of acute kidney injury. The current product information for Keppra informs that cases of acute kidney injury have been reported in patients treated with levetiracetam. Health Canada has requested that the other manufacturers of levetiracetam-containing products also update their product information with the same wording. More information is available [here](#).
- 01/05/2017 - The FDA has issued a new class warning for the use of benzodiazepines concomitantly with opiates. This would include AEDs such as **Onfi** (clobazam), **Klonopin** (clonazepam), **Ativan** (lorazepam), **Diastat** (rectal diazepam) and others. The warning states, "concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death." Please see the attached prescribing information regarding Onfi (clobazam)." More information [here](#) and [here](#). Find Onfi prescribing information [here](#).

2016 FDA Alerts

- 12/12/2016 - Health Canada carried out a safety review after learning that the European Medicines Agency was looking into a **potential interaction between levetiracetam and methotrexate**. The interaction between the two drugs may lead to higher amount of methotrexate in the blood, which may cause serious side effects. The side effects, which can be fatal, include sudden (acute) kidney failure.
- 11/18/2016 - FDA recalls of **Clonazepam** Tablets, USP, 0.5 mg, a drug used to treat certain seizure and panic disorders. The recall affects certain packages. More information [here](#).
- 11/7/2016 - Health Canada has indicated a potential link to respiratory depression—including sedation, somnolence, and loss of consciousness—in patients treated with **gabapentin**, particularly in patients having lung, kidney, or nervous system diseases that impact breathing, as well as in those that are elderly, or using other drugs that can affect breathing such as opioids, or in those are at higher risk of serious breathing problems. Additional information concerning the risk of respiratory depression has been added to the *Warnings and Precautions* section of the Canadian product monograph (CPM) for **Neurontin** (gabapentin). Concomitant use of CNS depressants with gabapentin is also a contributing factor. More information [here](#).
- 10/18/2016 - EMA recommends measures to ensure safe use of **Keppra** oral solution. More information [here](#).
- 10/17/2016 - FDA approves Carnexiv - i.v. **carbamazepine**, marketed by Lundbeck. More information [here](#).
- 10/6/2016 - FDA recalls **lamotrigine** tablets, USP, 150 mg, due to misprinted tablets. The medication (NDC 29300-113-05) was packaged in 500-count bottles. More information [here](#).
- 9/26/2016 - **Zarontin** (ethosuximide) - Defective Capsules Associated with Reduced Efficacy (Increased Frequency of Seizures). More information [here](#).
- 8/31/2016 - **Levetiracetam** may rarely be associated with acute renal failure. Recently, the Ministry of Health, Labour, and Welfare in Japan reported that a patient treated with levetiracetam developed

acute renal failure within 90 days of the initiation of the treatment. The condition resolved after discontinuation of levetiracetam. In Japan, the prescribing information for levetiracetam now lists acute renal failure as one of the serious adverse reactions. Currently, no specific monitoring is recommended with levetiracetam, but obtaining baseline renal tests before initiation of levetiracetam and periodically thereafter could be done to assess renal function.

- 8/30/2016 - **Lamotrigine** Orally Disintegrating Tablet 200 mg by Impax: Recall - Incorrect Labeling of Blister Cards. More information [here](#).
- 8/16/2016 - GlaxoSmithKline (GSK) is advising healthcare providers that **POTIGA** (ezogabine) Tablets, CV, 50mg, 200mg, 300mg, and 400mg will no longer be commercially available after June 30, 2017. GSK intends to permanently discontinue the product due to the very limited usage of the medicine and the continued decline in new patient initiation. Discontinuation is not due to efficacy or safety reasons. [Read more](#). [PDF]
- 8/15/2016 - In an enforcement letter the week of August 1st, the FDA ordered a recall of **Divalproex** 100 mg delayed-release tablets manufactured by Teva, Inc. due to failed specifications. The recall number is D-1449-2016 and code information is Lot #: 02D163, Exp. 9/2017.
- 5/29/2016 - A new boxed warning has been added to the prescribing information for ezogabine: "**Potiga** (ezogabine) can cause retinal abnormalities with funduscopy features similar to those seen in retinal pigment dystrophies, which are known to result in damage to the photoreceptors and vision loss. Macular abnormalities characterized as vitelliform lesions have also been observed. All patients taking Potiga should have baseline and periodic (every 6 months) systematic visual monitoring by an ophthalmic professional. Testing should include visual acuity, dilated fundus photography, and optical coherence tomography." [Click here to download](#). [PDF]
- 4/18/2016 - Study in *Neurology* finds **pregabalin** may be linked to major birth defects. [View the Pubmed citation](#).
- 4/12/2016 - The FDA issues overview, "FDA Regulation of Marijuana: Past Actions, Future Plans." [Click here to](#)

[download](#). [PDF]

- 4/6/2016 - The FDA and the manufacturer updated the prescribing information for **Zonegran** (zonisamide) to include a warning about Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) also known as multiorgan hypersensitivity. [Download the revised prescribing information here](#). [PDF]
- 2/19/2016 - FDA approves **Briviact** (brivaracetam) as an add-on treatment to other medications to treat partial-onset seizures in patients age 16 years and older with epilepsy. [Click here for more information](#).